



Assignment 10: Working with Files in Python



Instructions

1. Write each program in a separate Python file.
 2. Create the required text files in the same folder as your Python program.
 3. Display clear output messages.
 4. Close the file after use (or use the `with` statement).
 5. Test each program before submission.
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1 Create and Write to a File

Problem:

Write a Python program to create a file named `student.txt` and write the following text into it:

```
Name: Ahmed  
Class: 10  
Grade: A
```

Display a message after writing the data successfully.

Example Output

```
Data written successfully.
```



2 Read and Display File Contents

Problem:

Write a Python program to read and display all contents of the file `student.txt`.

Example Output

```
Name: Ahmed  
Class: 10  
Grade: A
```



3 Append Data to a File

Problem:

Write a Python program to add the following line to `student.txt` without deleting existing content:

```
School: ABC Public School
```

Display the updated contents of the file.

Example Output

```
Name: Ahmed  
Class: 10  
Grade: A  
School: ABC Public School
```

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2
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4 Count the Number of Lines in a File

Problem:

Write a Python program to count and display the total number of lines present in `student.txt`.

Example Output

```
Total Lines = 4
```

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5 Count the Number of Words in a File

Problem:

Write a Python program to count the total number of words present in `student.txt`.

Example Output

```
Total Words = 10
```

(The actual count may vary depending on the file content.)

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6 Copy Contents from One File to Another

Problem:

Write a Python program to copy all contents from `student.txt` to a new file named `backup.txt`.

Display a success message after copying.

Example Output

```
File copied successfully.
```

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Search for a Word in a File

Problem:

Write a Python program to check whether the word "Ahmed" exists in `student.txt`.

Example Output

```
Word found in the file.
```

or

```
Word not found in the file.
```

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Store Multiple Student Names in a File

Problem:

Write a Python program to store the following names in a file named `students.txt`:

```
Ahmed  
Fatima  
Rahul  
Aisha  
Zeeshan
```

Then read and display all names from the file.

Example Output

```
Ahmed  
Fatima  
Rahul
```

Aisha
Zeeshan

Bonus Question

Student Record System

Problem:

Write a program that:

1. Asks the user to enter a student's name and marks.
2. Stores the information in a file named `records.txt`.
3. Reads and displays all records stored in the file.

Example Output

```
Enter Name: Ahmed
Enter Marks: 85

Record saved successfully.

Ahmed - 85
```

Learning Outcomes

After completing this assignment, students will be able to:

✓ Create text files ✓ Write data to files ✓ Read data from files ✓ Append data to existing files ✓ Count lines and words in files ✓ Search data inside files ✓ Copy contents between files ✓ Build simple file-based applications using Python